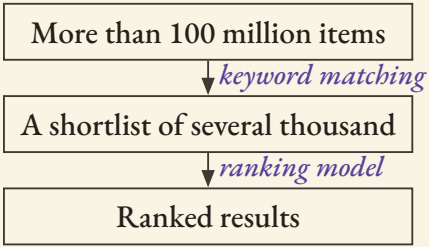
Etsy: a shortlist out of 100 million items

Etsy is a marketplace for handmade and vintage items.

Its shortlist used to come from keyword matching only, which fails when buyers and sellers use different words.

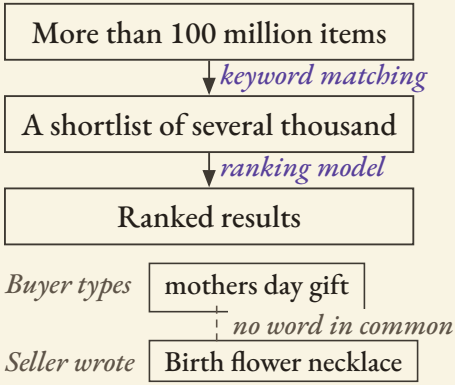


Jha and others (Etsy), IEEE FMLDS 2024. The listing is invented.

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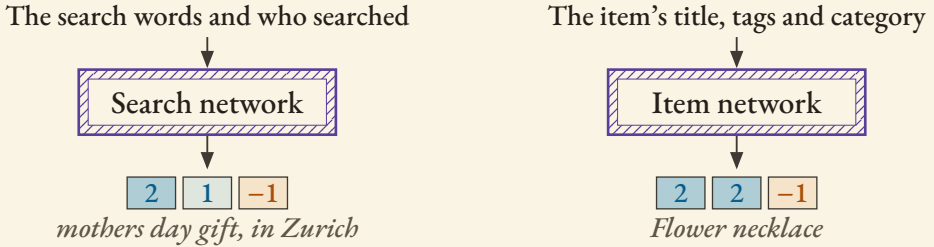
The training data: a month of search logs

1 month: more than 150 million rows, from up to 30 million users.

Search	Who searched	Item	Action
mothers day gift	Zurich, searched necklaces	Flower necklace	purchase
bridge photos	New York, clicked print shops	Bridge print	cart add
wool socks	Oslo, searched knitting	Knitted socks	cart add

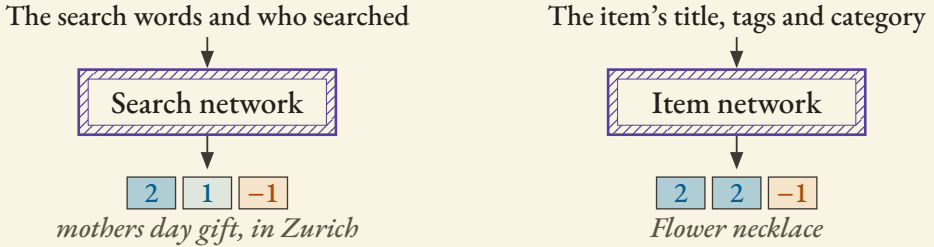
The rows are invented. Jha and others (Etsy), IEEE FMLDS 2024.

Every search and every item becomes a list of numbers



Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.

Every search and every item becomes a list of numbers



Training makes the inner product high for items bought or added to the cart after the search: here 7.

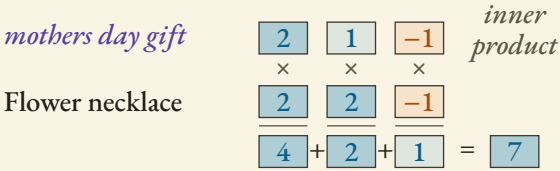
Invented numbers. Jha and others (Etsy), IEEE FMLDS 2024.

A search arrives: sort by inner product

mothers day gift 2 1 −1

Invented numbers.

A search arrives: sort by inner product



Invented numbers.

A search arrives: sort by inner product

				inner product
mothers day gift	2	1	−1	
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7
Gift wrap	−1	0	2	−4
Flower necklace	2	2	−1	7
Photo mug	1	1	0	3
Recipe board	1	2	0	4

Invented numbers.

A search arrives: sort by inner product

				inner product
mothers day gift	2	1	−1	
	×	×	×	
Flower necklace	2	2	−1	
	4	+ 2	+ 1	= 7
Gift wrap	−1	0	2	−4
Flower necklace	2	2	−1	7
Photo mug	1	1	0	3
Recipe board	1	2	0	4

Invented numbers.

Sorted: the top two join the shortlist

Flower necklace	7
Recipe board	4
Photo mug	3
Gift wrap	−4

Cost and value

Cost

72 hours *on 4 GPUs*

1 month *of search logs*

Value: A/B tests, relative change

+2.63% *site conversion rate*

+5.58% *search purchase rate*

Jha and others (Etsy), IEEE FMLDS 2024.

Cost and value

Cost

72 hours *on 4 GPUs*

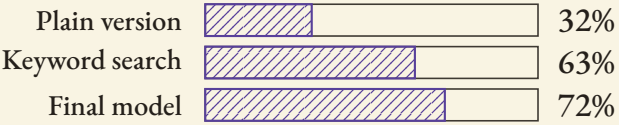
1 month *of search logs*

Value: A/B tests, relative change

+2.63% *site conversion rate*

+5.58% *search purchase rate*

Popular searches: share of the bought products found in the top 100



Jha and others (Etsy), IEEE FMLDS 2024.